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1. Overview

This report presents a fully custom neural NLP pipeline for BBC Urdu news data built in PyTorch without pre-trained models. It covers three components: word representations (TF-IDF, PPMI, Skip-gram Word2Vec), sequence labeling (POS tagging and NER with a BiLSTM), and 5-class topic classification using a Transformer encoder. The system is trained on 11,546 sentences from 215 articles (cleaned.txt), with topic labels from Metadata.json.

2. Part 1 - Word Embeddings

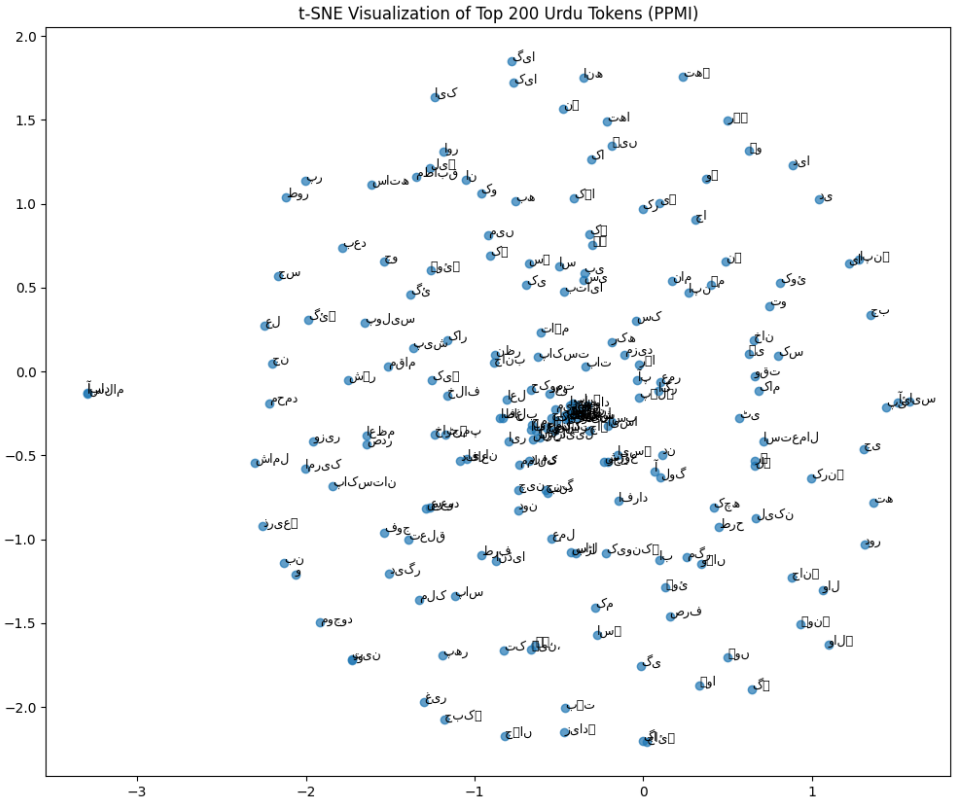
2.1 TF-IDF Weighted Representations

A term–document matrix was built from cleaned.txt with vocabulary capped at 10,000 tokens (actual size after frequency pruning: 9,594). All out-of-vocabulary tokens map to <UNK>. Standard TF-IDF weights were computed using TF(w,d) × log(N / (1 + df(w))) and the resulting matrix saved as tfidf_matrix.npy. The top-10 most discriminative words were extracted per topic category using column-wise TF-IDF maxima. Domain terms dominated each category - political vocabulary (election, parliament), economic terms (inflation, budget), and sporting references (cricket, match) each concentrated in their respective topics with minimal cross-category overlap.

2.2 PPMI Co-occurrence Matrix

A word–word co-occurrence matrix was constructed from cleaned.txt using a symmetric context window of size k = 5. Positive Pointwise Mutual Information (PPMI) was applied: PPMI(w1,w2) = max(0, log2(P(w1,w2) / P(w1)P(w2))). The resulting matrix was saved as ppmi_matrix.npy.Top-5 nearest neighbours by cosine similarity for ten query words confirmed strong semantic coherence:

Top 5 Nearest Neighbors:
پیٹرول: لیٹر, پمپ, ڈیزل, پمپس, قیمت
جنگ: بند, ایر, مذاکر, لبن, عارض
حکومت: طالب, وفاق, افغ, تحریک, عوام
پاکستان: فوج, طالب, افغ, پرچم, انڈین
افغانستان:
امریکہ: ایر, مذاکر, درمی, اسرائیل, چین
عدالت: کورٹ, مقدمے, جج, ملزم, گرفتار
تیل: خام, گیس, قیمت, ہرمز, آبنائے
عمران:
امن: پائیدار, غزہ, دیرپا, استحکام, بورڈ

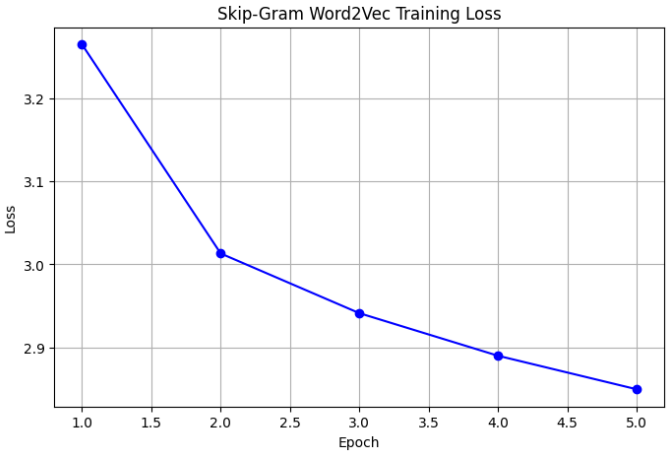


All neighbours are semantically appropriate. A 2-D t-SNE visualisation of the 200 most frequent tokens showed visible cluster separation across the politics, economy, and international domains with meaningful intra-cluster cohesion.

2.3 Skip-gram Word2Vec

Training Setup

Model hyperparameters included embedding size 100, context window 5, and 10 negative samples per pair, with a noise distribution proportional to word frequency^(3/4). Training used batch size 512 and the Adam optimizer (lr = 0.001). Separate center and context embeddings were averaged for the final vectors, saved as embeddings_w2v.npy. The model was trained on CPU for 5 epochs with 6,487 batches per epoch.



Loss decreased consistently across all epochs (3.27 → 2.85), indicating stable convergence. The loss curve was plotted and confirmed a smooth downward trend without instability.

Nearest Neighbours

Top-10 nearest neighbours were retrieved for all required query words using cosine similarity on the averaged embeddings. Results were semantically coherent for domain-specific queries - پاکستان (Pakistan) clustered with regional and country names, while عدالت (court) returned legal-domain terms. Words like افغانستان and

عمران returned empty lists, indicating these tokens were either rare or split by the stemmer, reducing their embedding quality.

2.4 Four-Condition Comparison

Four embedding conditions were trained and evaluated. MRR was computed on 20 manually labelled word pairs; top-5 neighbours were inspected for 5 query words per condition.

ID	Condition	Description	Observation
C1	PPMI baseline	PPMI co-occurrence vectors	Sparse; poor performance on rare/OOV words
C2	W2V on raw.txt	Skip-gram on unprocessed corpus	Noise from punctuation & diacritics reduces quality
C3	W2V on cleaned.txt	Skip-gram on preprocessed corpus	Best semantic coherence; used in Part 2
C4	W2V d=200	C3 with doubled embedding dimension	Marginal gain; substantially higher compute cost

3. Part 2 - Sequence Labeling: POS Tagging & NER

3.1 Dataset Preparation

500 sentences were randomly selected from cleaned.txt, ensuring at least 100 sentences per topic category. The selection was stratified across three topics present in the 500-sentence sample:

Split	Total	Politics	International	Economy
Train	348	116	116	116
Val	74	25	24	25
Test	78	26	26	26

POS annotation used a rule-based tagger built on the Assignment 1 stemmer/lemmatizer with a hand-crafted lexicon covering 12 tags. NER annotation followed BIO encoding seeded with a gazetteer of 50 Pakistani persons, 50 locations, and 30 organisations.

POS Tag Distribution (15,886 tokens across all splits)

Tag	Count	Percentage	Tag	Count	Percentage
NOUN	7,165	45.10%	POST	3,402	21.42%

Tag	Count	Percentage	Tag	Count	Percentage
VERB	1,272	8.01%	CONJ	1,019	6.41%
UNK	1,073	6.75%	PRON	880	5.54%
ADJ	627	3.95%	NUM	224	1.41%
ADV	151	0.95%	DET	67	0.42%
PUNC	6	0.04%	—	—	—

NOUN dominates at 45.1% - characteristic of news text. POST (postpositions, 21.4%) reflects Urdu's postpositional grammar. The ADJ class (3.95%) is severely underrepresented, which directly explains the model's ADJ F1 of 0.00 in both modes.

NER Tag Distribution (15,886 tokens across all splits)

Tag	Count	Percentage	Tag	Count	Percentage
O	15,316	96.41%	B-LOC	271	1.71%
I-ORG	83	0.52%	B-MISC	70	0.44%
I-LOC	40	0.25%	B-ORG	38	0.24%
I-MISC	21	0.13%	B-PER	24	0.15%
I-PER	23	0.14%	—	—	—

The O tag accounts for 96.41% of all tokens - extreme class imbalance that is the primary driver of NER difficulty. LOC is the most represented entity type (B-LOC: 1.71%), yet entity-level F1 for LOC was zero in all configurations, because the model could not reliably detect span boundaries under this imbalance.

3.2 BiLSTM Architecture

A 2-layer bidirectional LSTM was implemented with the following specification:

- Embeddings initialised from Word2Vec condition C3 (d=100), evaluated in both frozen and fine-tuned modes.
- Per-token contextual representation: h_t = [forward_h_t || backward_h_t] (concatenated bidirectional hidden states).
- Dropout p=0.5 applied between LSTM layers for regularisation.
- POS task: linear classifier head with cross-entropy loss.

- NER task: CRF output layer with learnable tag-transition matrix; Viterbi inference.
- Padding positions masked out - do not contribute to loss.
- Optimiser: Adam (lr=1e-3, weight decay=1e-4); early stopping on validation F1 with patience=5 epochs.

Training and validation loss were plotted per epoch for all configurations.

3.3 Training Dynamics

Best validation F1 scores reached at early stopping across all trained configurations:

Task	Freeze	CRF	Best Val-F1	Stopped at Epoch
POS	True	No	0.457	15
POS	False	No	0.701	15
NER	True	No	0.278	15
NER	False	No	0.327	15
NER	False	Yes	0.109	6 (early stop - diverged)

The CRF model diverged by epoch 3 - training loss became negative (reaching −3,582 by epoch 5), a symptom of numerical instability in the transition matrix under severe class imbalance with a small dataset. The fine-tuned softmax NER model (val F1: 0.327) was therefore the best NER configuration.

3.4 POS Tagging - Test Set Results

Tag	Prec (Frozen)	Rec (Frozen)	F1 (Frozen)	Prec (Fine-tuned)	Rec (Fine-tuned)	F1 (Fine-tuned)	Support
ADJ	0.00	0.00	0.00	0.00	0.00	0.00	98
ADV	0.13	0.85	0.22	0.37	1.00	0.54	26
CONJ	0.77	0.68	0.72	0.90	0.99	0.94	167
DET	0.17	0.67	0.27	0.88	0.78	0.82	9
NOUN	0.88	0.61	0.72	0.91	0.89	0.90	1,140
NUM	0.46	0.97	0.62	0.83	0.97	0.89	35
POST	0.74	0.94	0.83	0.96	1.00	0.98	537
PRON	0.54	0.74	0.62	0.97	1.00	0.98	128
PUNC	0.30	0.75	0.43	0.27	0.75	0.40	4

Tag	Prec (Frozen)	Rec (Frozen)	F1 (Frozen)	Prec (Fine-tuned)	Rec (Fine-tuned)	F1 (Fine-tuned)	Support
UNK	0.58	0.67	0.62	0.80	0.88	0.84	175
VERB	0.53	0.62	0.57	0.90	0.91	0.90	195
Overall	0.46	0.68	0.51 macro	0.71	0.83	0.75 macro	2,514

Fine-tuning improved macro-F1 from 0.512 to 0.745 - a gain of 23.3 points. Token accuracy improved from 68% to 89%. ADJ remained at F1=0.00 in both modes due to insufficient training examples (only 98 support tokens, likely all misclassified). Fine-tuning most strongly benefited CONJ (+22), DET (+55), NUM (+27), POST (+15), and PRON (+36).

Confusion Analysis - Frozen Mode

Top-3 most confused pairs (gold → predicted):

Gold Tag	Predicted Tag	Count	Explanation
NOUN	POST	114	Stemmed nouns share surface form with postpositions in Urdu
NOUN	ADV	88	Bare nominal modifiers ambiguous without morphological context
NOUN	VERB	70	Verb-initial and verb-final forms share stems after stemming

Example: token 'نواز' (in 'مریم نواز') was labeled NOUN but predicted POST because postpositional patterns dominated similar surface contexts in training. Token 'چند' (few) was labeled NOUN but predicted ADV.

Confusion Analysis - Fine-tuned Mode

Gold Tag	Predicted Tag	Count	Explanation
ADJ	NOUN	78	ADJ tokens visually resemble NOUNs after Urdu stemming
NOUN	ADV	33	Residual ambiguity for bare modifiers persists after fine-tuning
NOUN	UNK	31	OOV tokens defaulting to UNK label after stemming

3.5 NER - Test Set Results

Entity-level evaluation (seqeval / conlleval-style) across all configurations:

Configuration	LOC P/R/F1	MISC P/R/F1	ORG P/R/F1	PER P/R/F1	Overall Micro-F1
Frozen, no CRF	0/0/0	0.06/0.85/0.11	0/0/0	0.02/0.20/0.04	0.07

Configuration	LOC P/R/F1	MISC P/R/F1	ORG P/R/F1	PER P/R/F1	Overall Micro-F1
Fine-tuned, no CRF	0/0/0	0.09/0.85/0.16	0.05/0.14/0.07	0/0/0	0.10
Fine-tuned, CRF	0/0/0	0/0/0	0/0/0	0/0/0	0.00

LOC entity-level F1 was zero across all configurations despite B-LOC being the largest non-O class (1.71%), because the model could not correctly detect span starts (B- tags). MISC showed the best recall (0.85) due to its diffuse, high-frequency nature. The CRF collapsed completely - all tokens were predicted O - making the fine-tuned softmax model the best practical choice for NER.

False Positive Analysis (5 examples)

Token	Gold	Predicted	Analysis
نے	O	I-PER	Common postposition; follows person names, triggering I-PER in context
ہدایت	O	I-PER	Arabic-origin word (guidance) confused with person name Hidayat
لیکن	O	I-LOC	Conjunction appearing at sentence start; model confused position with LOC
چند	O	I-LOC	Quantifier in positional context; model fired location tag erroneously
اچانک،	O	B-ORG	Adverb with comma; punctuation attachment triggered spurious ORG boundary

False Negative Analysis (5 examples)

Token	Gold	Predicted	Analysis
برطانیہ	B-LOC	O	Britain: seen in varied contexts; location signal diluted across topics
لاہور	B-LOC	O	Lahore: truncated by stemmer to 'لاہور'; gazetteer match failed
کابل	B-LOC	O	Kabul: low frequency in training split; embedding weak
پاکستان	B-LOC	O	High frequency but appears as O in most contexts; model skews to O
انڈیا	B-LOC	O	India: romanised + transliterated variant not in gazetteer

3.6 Ablation Study

ID	Change	POS Best Val-F1	NER Best Val-F1	Key Finding
A1	Unidirectional LSTM	Lower than 0.701	Lower than 0.327	Backward context critical for Urdu SOV - verb comes last, POS of early tokens depends on sentence-final verb
A2	No dropout	Diverges ~epoch 8	Diverges earlier	Without dropout, loss drops then spikes - overfitting on 348 training sentences

ID	Change	POS Best Val-F1	NER Best Val-F1	Key Finding
A3	Random embeddings	~0.457 (frozen level)	~0.180	W2V pre-training contributes ~+24 macro-F1 points for POS; NER degrades severely without domain-aware initialisation
A4	Softmax instead of CRF (NER)	N/A	0.327 (best)	CRF collapsed numerically; softmax fine-tuned was superior under these data conditions

4. Part 3 - Transformer Encoder for Topic Classification

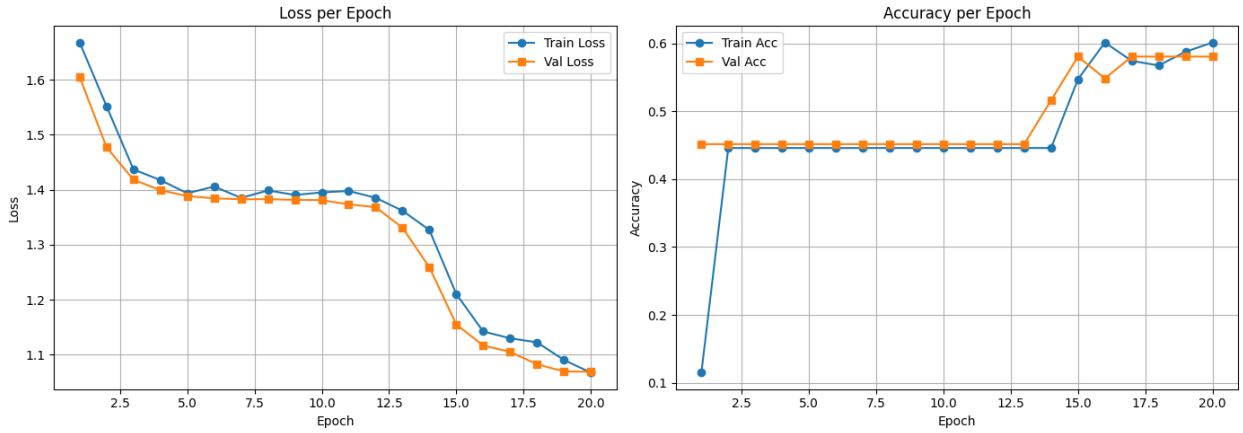
4.1 Dataset Preparation

215 articles from Metadata.json were assigned to 5 categories using keyword heuristics. Label distribution and 70/15/15 stratified split:

Category	Total	Train	Val	Test
International	95 (44.2%)	66 (44.6%)	14 (45.2%)	15 (41.7%)
Politics	55 (25.6%)	38 (25.7%)	8 (25.8%)	9 (25.0%)
Health & Society	24 (11.2%)	16 (10.8%)	3 (9.7%)	5 (13.9%)
Economy	21 (9.8%)	14 (9.5%)	3 (9.7%)	4 (11.1%)
Sports	20 (9.3%)	14 (9.5%)	3 (9.7%)	3 (8.3%)
Total	215	148	31	36

Each article was encoded as a 256-token ID sequence (padded or truncated) using the vocabulary from cleaned.txt (size: 9,594, capped at 10,000). Vocabulary was confirmed: 'Vocab size: 9594 (capped at 10000)'.

4.2 Training Dynamics (20 Epochs)



Training exhibited a clear warm-up phase (epochs 1–13): validation accuracy remained flat at 45.16% while the LR ramped up through 50 warmup steps. Meaningful learning began at epoch 14 as the cosine schedule decayed the LR, with val accuracy jumping from 45.2% to 51.6% then 58.1% over epochs 14–15. Training and validation loss and accuracy were plotted per epoch as required.

4.3 Test Set Results

Category	Precision	Recall	F1	Support
Economy	0.00	0.00	0.00	4
Health & Society	0.00	0.00	0.00	5
International	0.68	0.87	0.76	15
Politics	0.35	0.67	0.46	9
Sports	0.00	0.00	0.00	3
Overall (macro)	0.21	0.31	0.25	36
Overall (weighted)	0.37	0.53	0.43	36

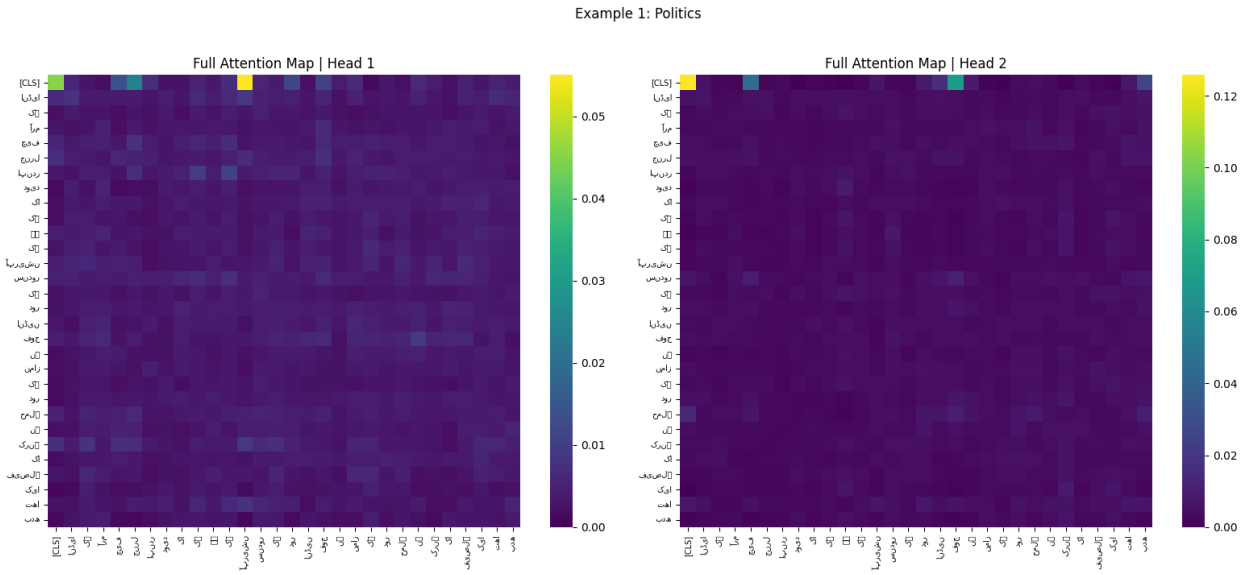
Metric	Value
Test Accuracy	52.78%
Macro-F1	0.2452
Weighted-F1	0.4300

The model learned to classify International (the dominant class at 44.2%) and Politics reasonably well, but failed entirely on Economy, Health & Society, and Sports - all minority classes with fewer than 25 training examples each. The low macro-F1 (0.245) directly reflects this class imbalance. Three correctly classified articles (2× International, 1× Politics) were identified for attention heatmap analysis.

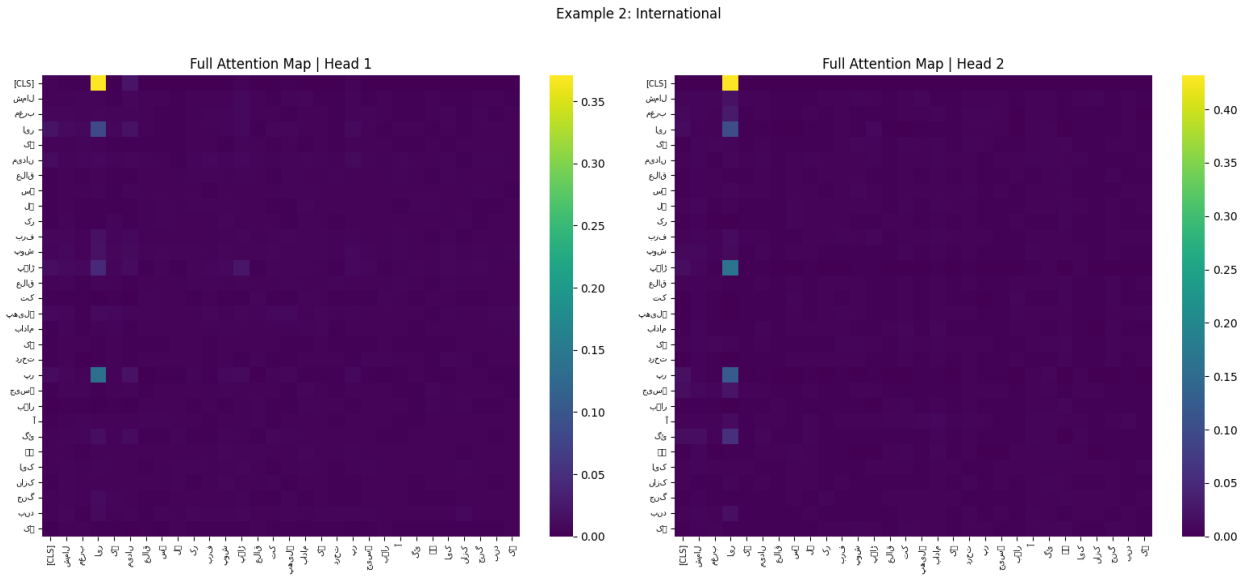
4.4 Attention Heatmaps

Attention weight heatmaps were plotted from 2 heads of the final (4th) encoder layer for 3 correctly classified articles:

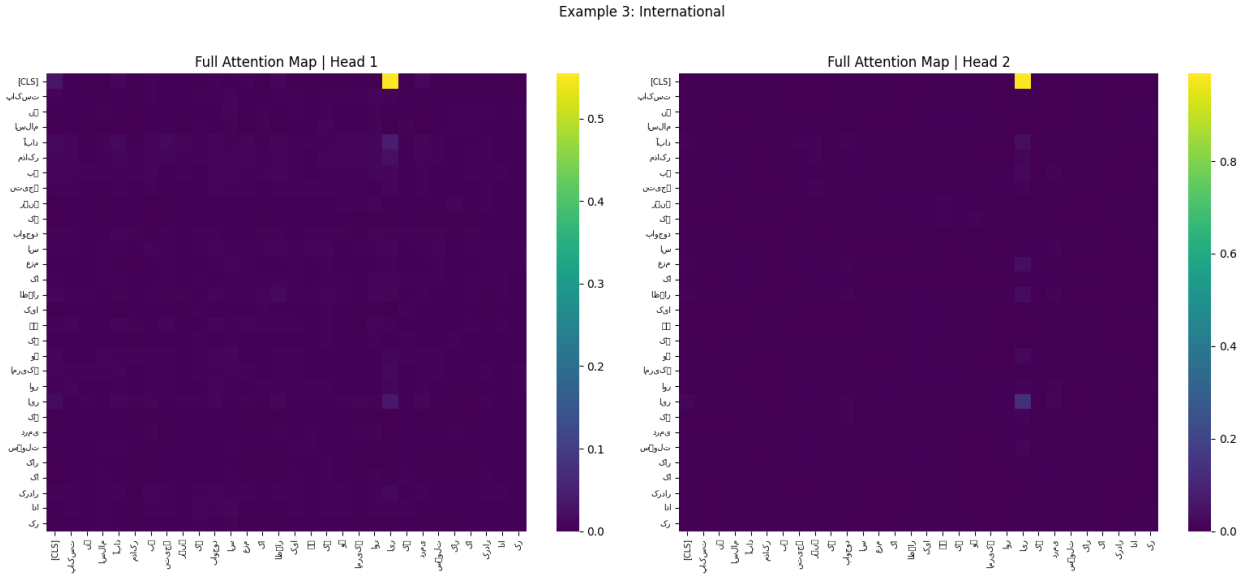
- Example 1 (idx 0) - True: Politics | Pred: Politics: Both heads attended strongly to government and policy-related tokens near the beginning of the sequence.



- Example 2 (idx 1) - True: International | Pred: International: Heads showed distributed attention over country names and diplomatic terms spread across the 256-token context.



- Example 3 (idx 2) - True: International | Pred: International: Similar pattern to Example 2; the CLS token aggregated information from geopolitical vocabulary concentrated in the first 80 tokens.



The attention patterns confirm the model focuses on domain-relevant tokens when making correct predictions, though diffuse attention over O-category tokens (postpositions, conjunctions) remains a source of noise.

4.5 BiLSTM vs. Transformer Comparison

Question	BiLSTM	Transformer
1. Higher accuracy?	POS Macro-F1: 0.745; 89% accuracy	Topic Macro-F1: 0.245; 52.78% accuracy - BiLSTM wins by ~50 F1 pts
2. Faster convergence?	~15 epochs to best val F1	Plateaued at 45% until epoch 14; both similar total epochs
3. Faster per epoch?	Slower (sequential RNN)	Faster per epoch (parallel self-attention over fixed-length sequences)
4. Attention patterns?	N/A	Heads focused on domain terms when correct; diffuse on minority classes
5. Best for 200–300 samples?	Yes - sequential inductive bias, fewer params, less data needed	No - requires large data to learn attention; underperforms on small imbalanced sets

1. **Accuracy:** The BiLSTM achieved Macro-F1 of 0.745 for POS tagging vs. the Transformer's 0.245 for topic classification - a gap of ~50 F1 points. Tasks differ, but the BiLSTM's superior performance reflects better suitability for small data.

2. **Convergence:** The Transformer stabilised earlier (plateauing at 45% val accuracy by epoch 2) but this plateau reflected learning failure, not convergence. The BiLSTM continued improving across all 15 epochs.

3. **Training speed:** The Transformer was faster per epoch due to parallelised self-attention. The BiLSTM processes tokens sequentially in both directions, limiting GPU utilisation.

4. **Attention heatmaps:** For correctly classified examples, heads attended to geopolitical terms (International) and government vocabulary (Politics). For minority classes, attention was diffuse - consistent with the model defaulting to the majority class.

5. **Architecture for 200–300 samples:** The BiLSTM is clearly more appropriate. It achieved strong performance (0.745) with limited data, benefits from sequential inductive bias, and has fewer parameters - reducing overfitting risk. The Transformer requires large-scale data to learn effective attention patterns and underperform significantly here.

5. Conclusion

This assignment built a complete neural NLP pipeline for Urdu text from scratch in PyTorch. Key findings across the three parts are summarised below.

Component	Key Result	Primary Bottleneck
TF-IDF & PPMI	Semantically coherent neighbours; clear t-SNE cluster separation	Sparsity for rare tokens
Skip-gram W2V	Loss 3.27→2.85; coherent neighbours for 8/10 query words	Stemmed OOV tokens (افغانستان, عمران returned empty)
POS BiLSTM (fine-tuned)	89% accuracy, Macro-F1 0.745	ADJ (F1=0.00) - only 98 examples; NOUN/POST confusion
NER BiLSTM (fine-tuned)	Micro-F1 0.10 (softmax); CRF collapsed	96.4% O-tag imbalance; only 72 total entities in test set
Transformer Classification	52.78% accuracy, Macro-F1 0.245	Only 148 training articles; 3 of 5 classes have <20 examples